Thermodynamic and structural properties of the thin radiating surface layer of a white dwarf

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We determine thermodynamic and structural properties of the Coulomb plasma (jellium) in the thin surface layer of a white dwarf using the path integral Monte Carlo method within a general relativity description of the star.

Keywords: White dwarf;Jellium;Path Integral;Monte Carlo;General Relativity;Thin surface layer;Thermodynamics;Structure

CONTENTS

I. INTRODUCTION

White dwarfs [1] are the dense remnants of stars, composed mainly of carbon and oxygen nuclei, which contain electrons, protons and neutrons and are supported against collapse induced by gravity by the electron degeneracy pressure due to Pauli exclusion principle. They remain white dwarfs till gravity overcomes electron degeneracy pressure, crushing electrons into protons to form neutrons through β^+ decay.

Therefore the simplest model for the interior of a white dwarf is that of *Jellium*. This is a system of pointwise electrons immersed in a uniform neutralizing background. The density of the electron gas is so high that the Fermi temperature is much higher than the temperature of the star and the electron gas can be considered degenerate. Chandrasekar [2–5] model of a white dwarf equilibrium structure didn't take into account the effects of general relativity. First Tolman [6], and later Oppenheimer and Volkoff [7] proposed a way to treat the hydrodynamic equations for the equilibrium and stability of a star within the framework of general relativity.

The luminosity of a white dwarf is due to a thin radiating layer on the star surface (see chapter 4 of the book [1]). The mechanism of a white dwarf cooling is the primary bridge between our theoretical description of the star and observational data which rely on essentially two aspects: luminosity observations and time scale of the observed cooling.

Purpose of this work is to determine the properties of the electron plasma in this thin surface layer using the path integral Monte Carlo (PIMC) method [8] at low non zero temperature. In order to take into account the effects of general relativity we will use the $3+1$ foliation of Arnowitt, Deser, and Misner (ADM) [9] to build a density matrix

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from the stress-energy tensor through a path integral where on each imaginary timeslice one selects a constant time foil [10]. This amounts to treat the quantum Coulomb plasma in the classical Schwarzschild spatial metric outside a spherical shell containing a mass $\mathscr{M}$ as confined in a thin spherical layer of width $\delta \ll \mathscr{R}$ at the surface of the star of radius $\mathscr{R}$. We will then approximate $\mathscr{M}$ as the white dwarf mass.

The problem of the plasma on the surface of the star, neglecting the radial thickness of the thin layer, has already been treated in Refs. [11–13]. Here we will consider the properties of the 'atmosphere' of a white dwarf.

We will also assume that the electrons dispersion relation is the one of non relativistic particles $\epsilon(p) = p^2/2m$. This is reasonable since the density of the electron-proton plasma [1] in the thin surface layer is $\rho_l \approx 10^3 \mathrm{g/cm}^3$ (see section 4.1 of the book [1]) much lower than the central densities. Therefore $p_F^2/m^2c^2 \approx 0.01$ where m is the electron mass and $p_F = h(3n/4\pi)^{1/3}$ is the polarized electrons Fermi momentum. The corresponding temperature of the surface layer fall in a range of temperatures $10^6\mathrm{K} \lesssim T \lesssim 10^7\mathrm{K}$. In the rest of this work we will use natural Planck units $c = G = \hbar = k_B = 1$ where c is the speed of light, $\hbar$ is the reduced Planck constant, G is the gravitational constant, and k_B is the Boltzmann constant.

II. ADM FOLIATION AND SCHWARZSCHILD METRIC

By Birkhoff theorem the vacuum solution of Einstein field equations outside a mass $\mathscr{M}$ is the one of Schwarzschild. Within the ADM foliation

$$||g_{\alpha\beta}|| = \begin{pmatrix} -(N^2 - N^i N_i) & N_i \\ N_i & g_{ij} \end{pmatrix}, \tag{2.1}$$

the *shift* $N_i = 0$, the *lapse* $N^2 = (1 - R_s/r)$, and $g_{rr} = (1 - R_s/r)^{-1}, g_{\theta\theta} = r^2, g_{\varphi\varphi} = r^2 \cos^2\theta$ with $R_s = 2\mathscr{M}$ the Schwarzschild radius. We will generally use a Greek index to denote one of the 4 spacetime components and a Roman index to denote one of the 3 spatial components. We will also introduce ${}^3g = \det\{g_{ij}\} = (1 - R_s/r)^{-1} r^4 \cos^2\theta > 0$ with $r > \mathscr{R} - \delta > R_s$.

Contracting Einstein field equations in vacuum implies a vanishing scalar curvature, the trace of Ricci tensor. Then Einstein field equations in vacuum reduce to the vanishing of the Ricci tensor. Therefore also the scalar curvature of a spatial foil, at constant time, of Schwarzschild metric must vanish. This will be used later on.

III. PIMC IN A FOIL OF SCHWARZSCHILD METRIC

A many body system composed of N *distinguishable* particles of mass m with positions in $R = (\boldsymbol{r}_1, \boldsymbol{r}_2, \ldots, \boldsymbol{r}_N) = (\{\boldsymbol{r}_l\})$ where each position vector $\boldsymbol{r}_i = (r_i^1, r_i^2, r_i^3) = (\{r_l^j\})$ in 3 dimensions. On a foil of Schwarzschild solution (2.1) with spatial metric tensor $g_{ij}(\boldsymbol{r})$, the geodesic distance between two infinitesimally close points R and R' is $d\tilde{s}^2(R, R') = \sum_{l=1}^N ds^2(\boldsymbol{r}_l, \boldsymbol{r}'_l)$ where $ds^2(\boldsymbol{r}, \boldsymbol{r}') = g_{ij}(\boldsymbol{r})(\boldsymbol{r} - \boldsymbol{r}')^i(\boldsymbol{r} - \boldsymbol{r}')^j$. Moreover,

$$\tilde{g}_{ij}(R) = g_{i_1 j_1}(\boldsymbol{r}_1) \otimes \ldots \otimes g_{i_N j_N}(\boldsymbol{r}_N), \tag{3.1}$$

$$\tilde{g}(R) = \prod_{l=1}^N {}^3g(\boldsymbol{r}_l), \tag{3.2}$$

where $||\tilde{g}_{ij}||$ is a matrix made of N diagonal blocks $||g_{i_l j_l}||$ with $l = 1, 2, \ldots, N$. The Laplace-Beltrami operator on the manifold of dimension $3N$ is

$$\Delta_R = \tilde{g}^{-1/2} \nabla_i (\tilde{g}^{1/2} \tilde{g}^{ij} \nabla_j), \tag{3.3}$$

where $\nabla = \nabla_R$, $\tilde{g}^{ij}$ is the inverse of $\tilde{g}_{ij}$, i.e. $\tilde{g}_{ij}\tilde{g}^{jk} = \delta_i{}^k$ the Kronecker delta.

The Hamiltonian $\mathscr{H}$ of the N non relativistic particles will be

$$\mathscr{H} = -\lambda \Delta_R + V(R), \tag{3.4}$$

where $\lambda = \hbar^2/2m$ and $V(R)$ is the *physical potential* energy of the system, that we here assume only a function of the particles positions and bounded from below.

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[1] Assuming that the white dwarf is made up of a completely ionized plasma then we may write $\rho = \mu_e m_u n$ with n the electrons number density, m_u the atomic mass unit, and $\mu_e = A/Z$ with Z the atomic number and A the mass number, so that for example for a ^{4}He or a ^{12}C white dwarf $\mu_e = 2$ and for a ^{56}Fe white dwarf $\mu_e = 56/26 \approx 2.134$.

The density matrix ρ of the system obeys to Bloch equation

$$\frac{\partial\rho(t)}{\partial t} = -\mathscr{H}\rho(t), \tag{3.5}$$

$$\rho(0) = 1\!\!1, \tag{3.6}$$

where t is the imaginary time with dimension of the reciprocal of energy and $1\!\!1$ the identity matrix. The position representation of the density matrix is then obtained from $\rho(R,R';t) = \langle R|\rho(t)|R'\rangle$ with $\langle R|R'\rangle = \delta(R-R')/\sqrt{\tilde{g}(R)}$ where δ is a $3N$ dimensional Dirac delta function. In the small imaginary time τ limit the position representation of the density matrix is [13]

$$\rho(R,R';\tau) \propto \tilde{g}(R)^{-1/4}\sqrt{\mathscr{D}(R,R';\tau)}\,\tilde{g}(R')^{-1/4}e^{\lambda\tau\mathscr{R}(R)/6}e^{-\mathscr{S}(R,R';\tau)}, \tag{3.7}$$

where $\mathscr{R} = 0$ is the scalar curvature of the Schwarzschild spatial foil [2], $\mathscr{S}$ the action, and $\mathscr{D}$ the van Vleck's determinant [15, 16]

$$\mathscr{D}_{\mu\nu} = -\nabla_\mu\nabla'_\nu\mathscr{S}(R,R';\tau), \tag{3.8}$$

$$\det||\mathscr{D}_{\mu\nu}|| = \mathscr{D}(R,R';\tau), \tag{3.9}$$

where $\nabla = \nabla_R$ and $\nabla' = \nabla_{R'}$. This determinant is the Jacobian of the transformation from the initial conditions given by fixing the pair of momentum and coordinate of each particle to the boundary conditions given by specifying the pair of initial and final coordinates needed in the coordinate representation. The square root of the van Vleck determinant factor takes into account the density of paths among the minimum extremal region for the action (see Chapter 12 of Ref. [16]). The two factors $\tilde{g}^{-1/4}$ are needed in order to have for the density matrix a bidensity for which the boundary condition to Bloch equation is simply a Dirac delta function. In the small $\tau \to 0$ limit [3] $\tilde{g}(R)^{-1/4}\sqrt{D(R,R';\tau)}\,\tilde{g}(R')^{-1/4} \to (1/2\lambda\tau)^N$.

For the *action* $\mathscr{S}$, the *kinetic-action* $\mathscr{K}$, and the *inter-action* $\mathscr{U}$ we have [4]

$$\mathscr{S}(R,R';\tau) = \mathscr{K}(R,R';\tau) + \mathscr{U}(R,R';\tau), \tag{3.10}$$

$$\mathscr{K}(R,R';\tau) = \frac{dN}{2}\ln(4\pi\lambda\tau) + \frac{d\tilde{s}^2(R,R')}{4\lambda\tau}. \tag{3.11}$$

In particular the kinetic-action is responsible for a diffusion of the random walk with a single particle variance equal to $\sigma_{ij}^2(\boldsymbol{r}) = 2\lambda\tau/g_{ij}(\boldsymbol{r})$. The inter-action is defined as $\mathscr{U} = \mathscr{S} - \mathscr{K}$ and for potential energies bounded from below one can resort to the Trotter formula [17] to reach the *primitive approximation* [5]

$$\mathscr{U}(R,R';\tau) = \tau[V(R) + V(R')]/2. \tag{3.12}$$

Given then an observable $\mathscr{O}$ we can determine its thermal average at an absolute temperature T from

$$\langle\mathscr{O}\rangle = g_s\mathrm{tr}\{\rho(\beta)\mathscr{O}\}/Z_N, \tag{3.13}$$

$$Z_N = g_s\mathrm{tr}\{\rho(\beta)\}, \tag{3.14}$$

where $\beta = 1/T$, $\mathrm{tr}\{\ldots\}$ is the trace over the spatial variables, Z_N the canonical partition function, $g_s = 2s+1$ is the spin degeneracy, and we assumed the Hamiltonian independent from spin. For *identical* bodies the spin-statistics theorem of quantum field theory, dictates that in spatial dimension bigger than two, particles with integer spin are bosons (obeying Bose-Einstein statistics, symmetric wavefunctions), while half-integer spin particles are fermions (obeying Fermi-Dirac statistics, antisymmetric wavefunctions). We will only consider polarized electrons that are fermions with $g_s = 1$.

The position representation of the density matrix at an imaginary time $t = \beta$ is obtained through a *path integral*

$$\rho(R,R';\beta) = \langle R|\rho(\beta)|R'\rangle = \int\prod_{k=1}^{M-1}\left[\rho(R_k,R_{k+1};\tau)\sqrt{\tilde{g}(R_k)}\,dR_k\right]\delta(R_0-R)\delta(R_M-R')\,dR_0dR_M, \tag{3.15}$$

where we have discretized the imaginary time β into M *timeslices* with a small *timestep* $\tau = \beta/M$. The volume element of integration, $\sqrt{\tilde{g}(R)}\,dR$, can be treated by introducing $W(R) = -\ln[\sqrt{\tilde{g}(R)}]/\tau$ as an effective *geometrical*

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[2] The factor depending on the curvature of the manifold is due to Bryce DeWitt [14]. For a space of constant curvature there is clearly no effect, as the term due to the curvature just leads to a constant multiplicative factor that has no influence on the measure of the various observables.

[3] Remember that the metric tensor is covariantly constant.

[4] The expression for $\mathscr{K}$ is the one of Eq. (24.16) of Ref. [16] to lowest order in $R-R'$.

[5] See Ref. [8] for a numerical analysis of the accuracy of this approximation and for its possible refinements.

potential. The many body path $R(t)$ is consequently discretized in M *beads* $R_k = (\{\boldsymbol{r}_{l,k}\}) = (\{r^j_{l,k}\})$ at each timeslice $k = 1, 2, \ldots, M$. We will also call *link* a pair of contiguous beads. Note that in order to measure an observable through Eq. (3.13) it is necessary to consider closed paths such that $R(t+\beta) = R(t)$, or rings on the foil.

For identical electrons, that satisfy to the Fermi-Dirac statistics, one needs to antisymmetrize the distinguishable density matrix (3.15) [18]. In these cases we can then write [6]

$$\rho_{\rm F}(R, R'; \beta) = \frac{1}{N!} \sum_{\mathscr{P}} \mathrm{sgn}(\mathscr{P}) \rho(\mathscr{P}R, R'; \beta), \tag{3.16}$$

$$\mathrm{sgn}(\mathscr{P}) = (-1)^{\sum_{\nu=1}^N (\nu-1) C_\nu}, \tag{3.17}$$

where $\mathscr{P}$ is any permutation of the N particles such that $\mathscr{P}R = (\boldsymbol{r}_{\mathscr{P}1}, \boldsymbol{r}_{\mathscr{P}2}, \ldots, \boldsymbol{r}_{\mathscr{P}N})$, with sign $\mathrm{sgn}(\mathscr{P})$. Any permutations can be broken into cycles $\mathscr{P} = \{C_\nu\}$ where C_ν is the number of cycles of length ν in $\mathscr{P}$. An even (odd) permutation has $\mathrm{sgn}(\mathscr{P}) = +1(-1)$ and an even (odd) number of exchanges of a pair of particles.

We will model the Coulomb plasma in the white dwarf atmosphere as a jellium, i.e. a system of N electrons immersed in a uniform neutralizing background. The physical potential energy is then

$$V(R) = \sum_m \phi_b(r_m) + \sum_{m<n} \phi(\boldsymbol{r}_m, \boldsymbol{r}_n), \tag{3.18}$$

with ϕ_b the potential energy due to a uniform proper charge density $\rho_b = -ne$ and ϕ the Coulomb pair potential between a charge q_m at $\boldsymbol{r}_m$ and a charge q_n at $\boldsymbol{r}_n$ and ϕ_b the potential energy of the uniform neutralizing background. The exact analytical expression is given by the Copson-Linet potential [19–21] [7]

$$\phi(\boldsymbol{r}_1, \boldsymbol{r}_2) = \frac{q_1 q_2}{r_1 r_2} \left[\frac{(r_1 - \mathscr{M})(r_2 - \mathscr{M}) - \mathscr{M}^2 \cos\gamma}{\sqrt{(r_1 - \mathscr{M})^2 + (r_2 - \mathscr{M})^2 - 2(r_1 - \mathscr{M})(r_2 - \mathscr{M})\cos\gamma - \mathscr{M}^2 \sin^2\gamma}} + \mathscr{M} \right], \tag{3.19}$$

where

$$\cos\gamma = \sin\theta_1 \sin\theta_2 + \cos\theta_1 \cos\theta_2 \cos(\varphi_1 - \varphi_2). \tag{3.20}$$

In the flat spacetime $\mathscr{M} \to 0$

$$\phi(\boldsymbol{r}_1, \boldsymbol{r}_2) = \frac{q_1 q_2}{\sqrt{r_1^2 + r_2^2 - 2 r_1 r_2 \cos\gamma}}. \tag{3.21}$$

The charge neutrality of the jellium in the spherical crown Ω such that $\mathscr{R} < r < \mathscr{R} - \delta$ requires

$$\rho_b = -\frac{Ne}{\frac{4}{3}\pi \left[\mathscr{R}^3 - (\mathscr{R} - \delta)^3\right]}, \tag{3.22}$$

with e the electron charge. The Poisson equation for the background potential energy ϕ_b is

$$\frac{1}{r^2} \frac{d}{dr} \left[r^2 \sqrt{1 - R_s/r}\, E \right] = 4\pi e \rho_b, \tag{3.23}$$

where $E(r) = -d\phi_b(r)/dr$ is the radial electric field. Integrating the Poisson equation from the origin $r = 0$ to r yields the internal electric field

$$E(r) = -\frac{4\pi e \rho_b}{3} \frac{r}{\sqrt{1 - R_s/r}}. \tag{3.24}$$

Integrating $E(r)$ yields the background potential profile

$$\phi_b(r) = \frac{\pi e \rho_b}{3} \left[r(2r + 3R_s)\sqrt{1 - R_s/r} + 3R_s^2 \mathrm{atanh}\sqrt{1 - R_s/r} \right],$$

up to an additive constant. Which reduces to the usual quadratic potential $\phi_b(r) \to 2\pi r^2 e \rho_b/3$ in the flat $R_s \to 0$ limit.

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[6] One can antisymmetrize respect to the first, the second, or both the arguments of the distinguishable density matrix. We here choose the first case.

[7] Compare with the one on a Flamm paraboloid [22].

IV. PIMC CONSTRAINED IN A RADIAL THIN SURFACE LAYER

From exercise 4.1 of the book [1] follows that the depth of the thin radiating surface layer of a white dwarf is

$$\delta \lesssim 10^{-2}\mathscr{R}, \tag{4.1}$$

where the typical radius of a white dwarf is $\mathscr{R} \approx 10^{-3}R_\odot - 10^{-2}R_\odot$.

In order to treat the problem of electrons constrained in a thin spherical shell of width δ we imagine an infinite repulsive external potential at $r > \mathscr{R}$ and at $r < \mathscr{R} - \delta$. In the Monte Carlo we reject any move that brings an electron in these two regions.

Of course this is a first brute approximation to a more realistic model including an ‘osmosis’ of electrons between the thin surface layer of the white dwarf atmosphere and the star interior.

The Cartesian coordinates of a particle on the Schwarzschild spatial foil is

$$\begin{cases} x = r\cos\theta\cos\varphi \\ y = r\cos\theta\sin\varphi \\ z = r\sin\theta \end{cases} \tag{4.2}$$

where $r^1 = r \in [\mathscr{R} - \delta, \mathscr{R}]$, $r^2 = \theta \in [-\pi/2, \pi/2[$ with $\theta = 0$ a Flamm paraboloid equatorial section of the spatial foil, $r^3 = \varphi \in [-\pi, \pi[$, and the particle path on it is $\boldsymbol{q}(t) = (x(t), y(t), z(t))$.

V. NUMERICAL RESULTS

We will choose $\mathscr{R} = 10^{-3}R_\odot$ with δ from Eq. (4.1), $\mathscr{M} = M_\odot$, and $T = 10^6\mathrm{K} - 10^7\mathrm{K}$.

We use the Metropolis algorithm [23, 24] to evaluate the average of Eq. (3.13).

In order to explore ergodically the positions space $\boldsymbol{r} = (r, \theta, \varphi)$ to sample the distinguishable density matrix we use the transition *displacement* move described in Appendix A of Ref. [13].

In order to sample the permutation sum of Eq. (3.16) needed for fermions we use a transition move combination of 2 Brownian *bridges* between unlike electrons as described in Appendix B of Ref. [13]. To construct a single Brownian bridge it is essential to map, project, the Schwarzschild spatial foil on a flat coordinate system. Our choice is presented in Appendix B of Ref. [13]. Perform the Gaussian bridge move in the projection flat space and then map it back to the Schwarzschild spatial foil. The Metropolis algorithm will then allow to sample the high temperature density matrix whose kinetic action is not purely Gaussian on the Schwarzschild spatial foil, due to the metric tensor appearing in $d\tilde{s}^2$.

We will work in the canonical ensemble with fixed number of particles N, electron number density $n = -\rho_b/e$, and absolute temperature $T = 1/k_B\beta$. The jellium *degeneracy parameter* is $\Theta = T/T_D$ where the degeneracy temperature $T_D = n^{2/3}\hbar^2/mk_B$ [8]. For temperatures higher than T_D, $\Theta \gg 1$, quantum effects are not very important. The *Coulomb coupling constant* is $\Gamma = \beta e^2/a_0 r_s$ with $a_0 = \hbar^2/me^2$ the Bohr radius and $r_s = (4\pi n/3)^{-1/3}/a_0$ the Wigner-Seitz radius. At weak coupling, $\Gamma \ll 1$, the plasma becomes weakly correlated. We then see that the kinetic energy scales as $1/r_s^2$ and the potential energy as $1/r_s$ so that the electron gas will approach the ideal gas limit at low r_s or high number density n.

In Table I we report two cases studied in our simulations at two different temperatures $T = 10^6\mathrm{K}$ in case A and $T = 10^7\mathrm{K}$ in case B. As we can see from the values of Γ and Θ the atmosphere is expected to be strongly quantum in case B and strongly correlated in case A. The timestep is kept fixed at $\tau = \beta/M = 1.4 \times 10^{24}$ in reduced units.

Apart from thermodynamic properties like the *kinetic energy* per particle e_K and the *potential energy* per particle e_V [9] we will also measure structural properties like the *radial distribution function*, $g(r) = \langle \mathscr{O} \rangle$. For it we used the following histogram estimator,

$$O(R; r, a) = \sum_{m \neq n} \frac{1_{]a-\Delta/2, a+\Delta/2]}(r_m^1) 1_{]a-\Delta/2, a+\Delta/2]}(r_n^1) 1_{]r-\Delta/2, r+\Delta/2]}(d_{mn})}{N n_{id}(r)}, \tag{5.1}$$

where Δ is the histogram bin, $1_{]u,v]}(x) = 1$ if $x \in]u, v]$ and 0 otherwise, d_{mn} is the Euclidean distance between 2 electrons m, n at the same fixed radial distance $a \in [\mathscr{R} - \delta, \mathscr{R}]$ from the center of the star

$$d_{mn} = d(\boldsymbol{r}_m, \boldsymbol{r}_n) = a\sqrt{2(1 - \hat{\boldsymbol{q}}_m \cdot \hat{\boldsymbol{q}}_n)} = 2a\sin[\arccos(\hat{\boldsymbol{q}}_m \cdot \hat{\boldsymbol{q}}_n)/2], \tag{5.2}$$

[8] This is the temperature at which the size of a single electron path, in absence of interaction, i.e. the thermal wavelength $(2\lambda\beta)^{1/2}$, equals the separation between single particle paths, i.e. roughly $n^{-1/3}$. Below this temperature it is possible for single particle paths to link up exchanging their end points.

[9] The estimators for these observables are carefully described in Ref. [8].

TABLE I. Cases treated in our simulations: $\mathscr{R}$ the white dwarf radius, $\delta = 10^{-2}\mathscr{R}$ the white dwarf atmosphere thickness, $\mathscr{M}$ the white dwarf mass, N the number of particles, r_s the Wigner-Seitz radius, β the inverse temperature, $e_K = \langle K\rangle/N$ the kinetic energy per particle from the thermodynamic estimator as explained in Ref. [8], $e_V = \langle V\rangle/N$ the physical potential energy per particle, and $e_W = \langle W\rangle/N$ the geometrical potential energy per particle. We chose natural Planck units with $c = G = \hbar = k_B = 1$. The adimensional quantities Γ and Θ were introduced in the main text. The simulation lasted more than 10^8 Monte Carlo steps where one step is made of a displace move of all the beads of a single particle path as described in Appendix A of Ref. [13] and a bridge move as described in Appendix B of Ref. [13].

case	M	N	$\mathscr{R}$	δ	$\mathscr{M}$	$a_0 r_s$	β	Γ	Θ	Ne_K	Ne_V	$-Ne_W$
A	100	30	$R_\odot 10^{-3}$	$R_\odot 10^{-5}$	$M_\odot$	6×10^{23}	1.4×10^{26}	17.7759	0.252952			
B	10	30	$R_\odot 10^{-3}$	$R_\odot 10^{-5}$	$M_\odot$	6×10^{23}	1.4×10^{25}	1.77759	2.52952			

and

$$n_{id}(r) = \sigma\pi\left[(r+\Delta/2)^2 - (r-\Delta/2)^2\right] \ , \tag{5.3}$$

is the average number of particles on the spherical crown $]r-\Delta/2, r+\Delta/2]$ of radius a for the ideal gas of surface density $\sigma = n\Delta$. We have that $\sigma^2 g(r)$ gives the probability, that sitting on a particle at $\boldsymbol{r}$, one finds another particle at $\boldsymbol{r}'$ with $r = d(\boldsymbol{r},\boldsymbol{r}') \in [0, 2a]$. Note that $\sqrt{2a^2}$ is the Euclidean distance between a pole and a point on the equator.

Fermions properties cannot be calculated exactly with path integral Monte Carlo because of the *fermions sign problem* [18, 25]. We then have to resort to an approximate calculation. The one we chose was the Restricted Path Integral (RPIMC) approximation [11, 18, 25, 26] with a "free fermions restriction". The trial density matrix used in the *restriction* is chosen as the one reducing to the ideal density matrix in the limit of $t \ll 1$ and is given by the following explicit analytic expression,

$$\rho_0(R', R; t) \propto \det\left\|e^{-\frac{d^2(\boldsymbol{r}_i', \boldsymbol{r}_j)}{4\lambda t}}\right\| . \tag{5.4}$$

Note that here we purposely chose the Euclidean distance

$$d(\boldsymbol{r}', \boldsymbol{r}) = \sqrt{(\boldsymbol{q}-\boldsymbol{q}')\cdot(\boldsymbol{q}-\boldsymbol{q}')}. \tag{5.5}$$

We did this just for the sake of simplicity. In any case both the Euclidean and the geodesic distance reduce to $ds^2(\boldsymbol{r}',\boldsymbol{r}) = g_{\alpha\beta}(\boldsymbol{r})(\boldsymbol{r}-\boldsymbol{r}')^\alpha(\boldsymbol{r}-\boldsymbol{r}')^\beta$ in the $t \ll 1$ limit. The implementation of the RPIMC on a curved space has already been described in Ref. [13].

VI. CONCLUSIONS

AUTHOR DECLARATIONS

Conflicts of interest

None declared.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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